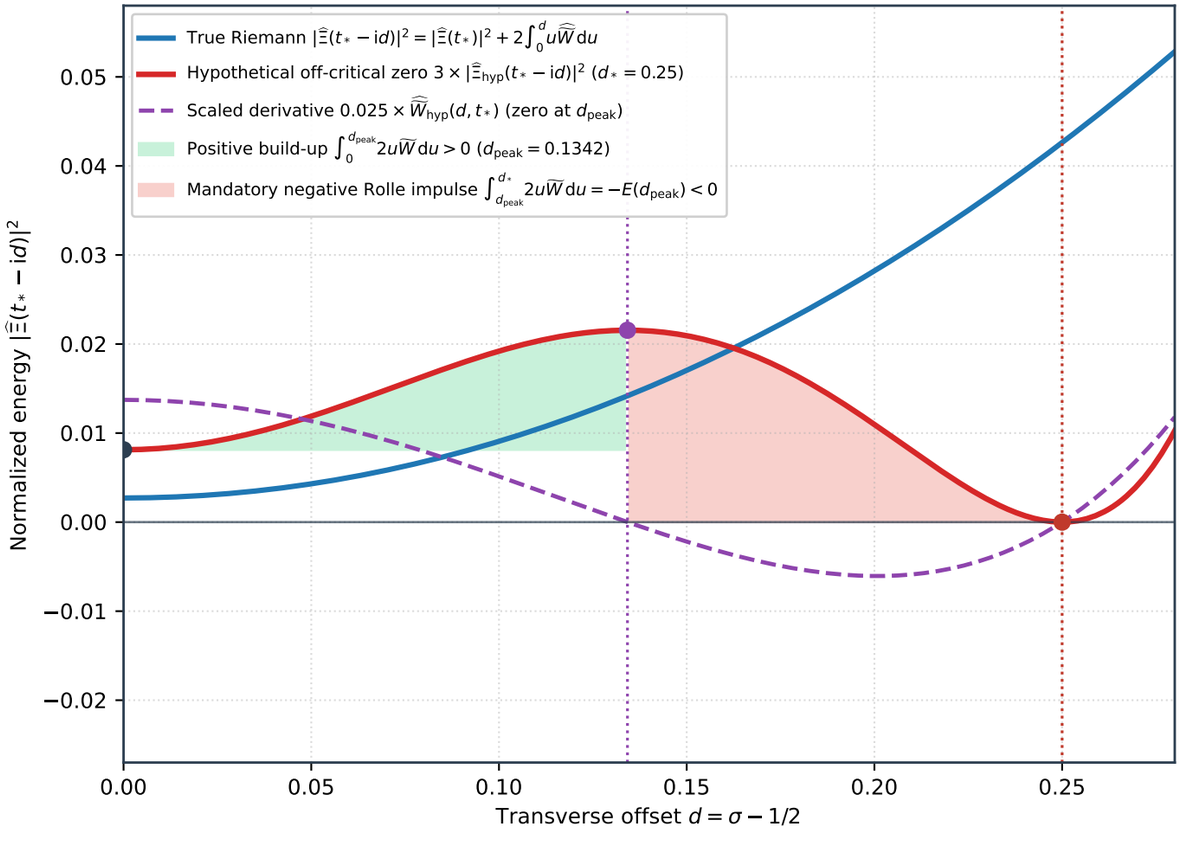
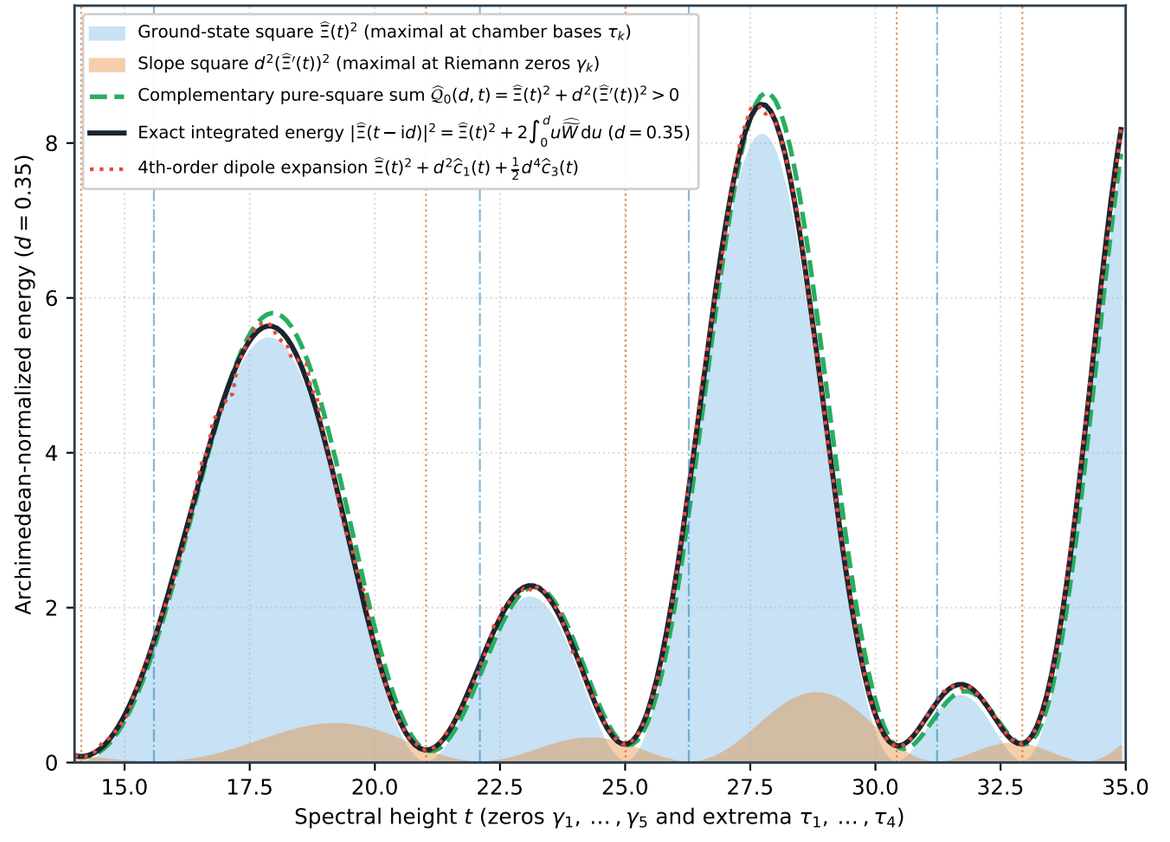
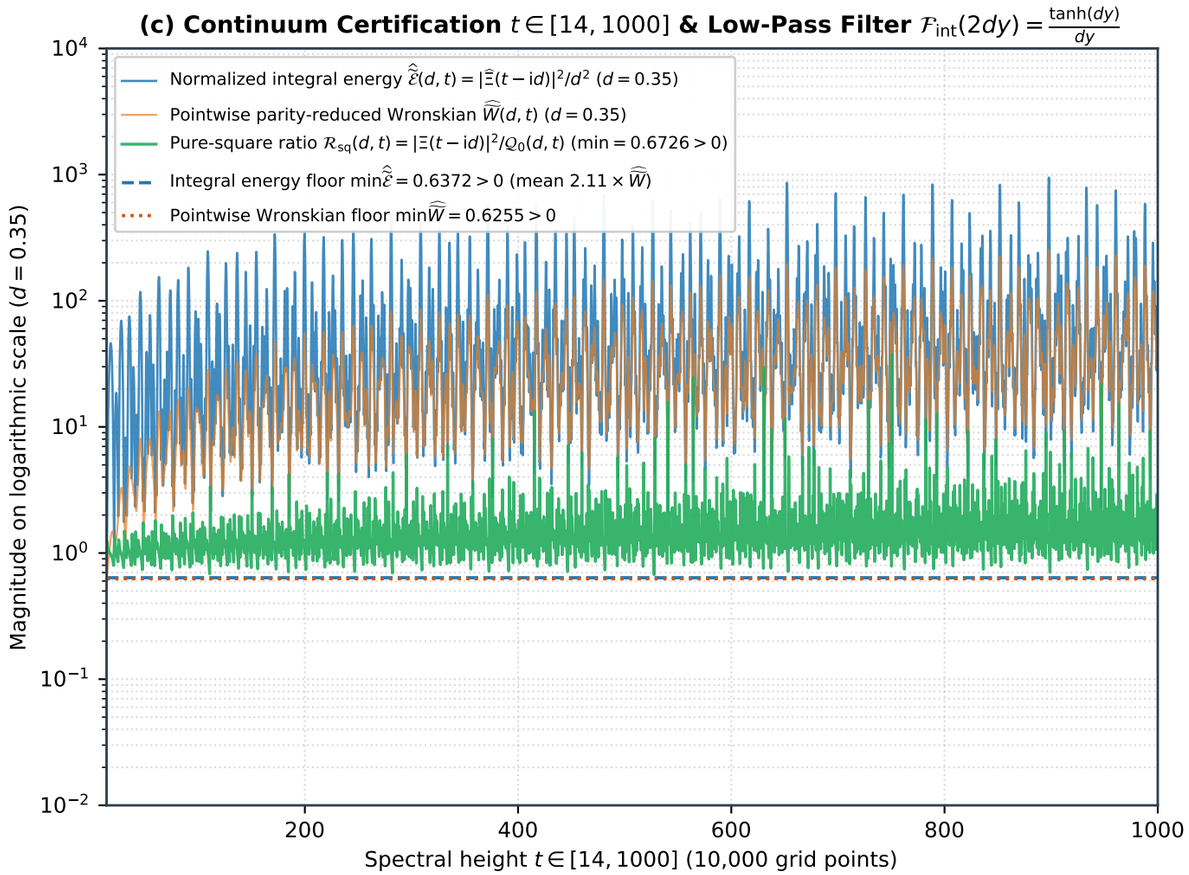
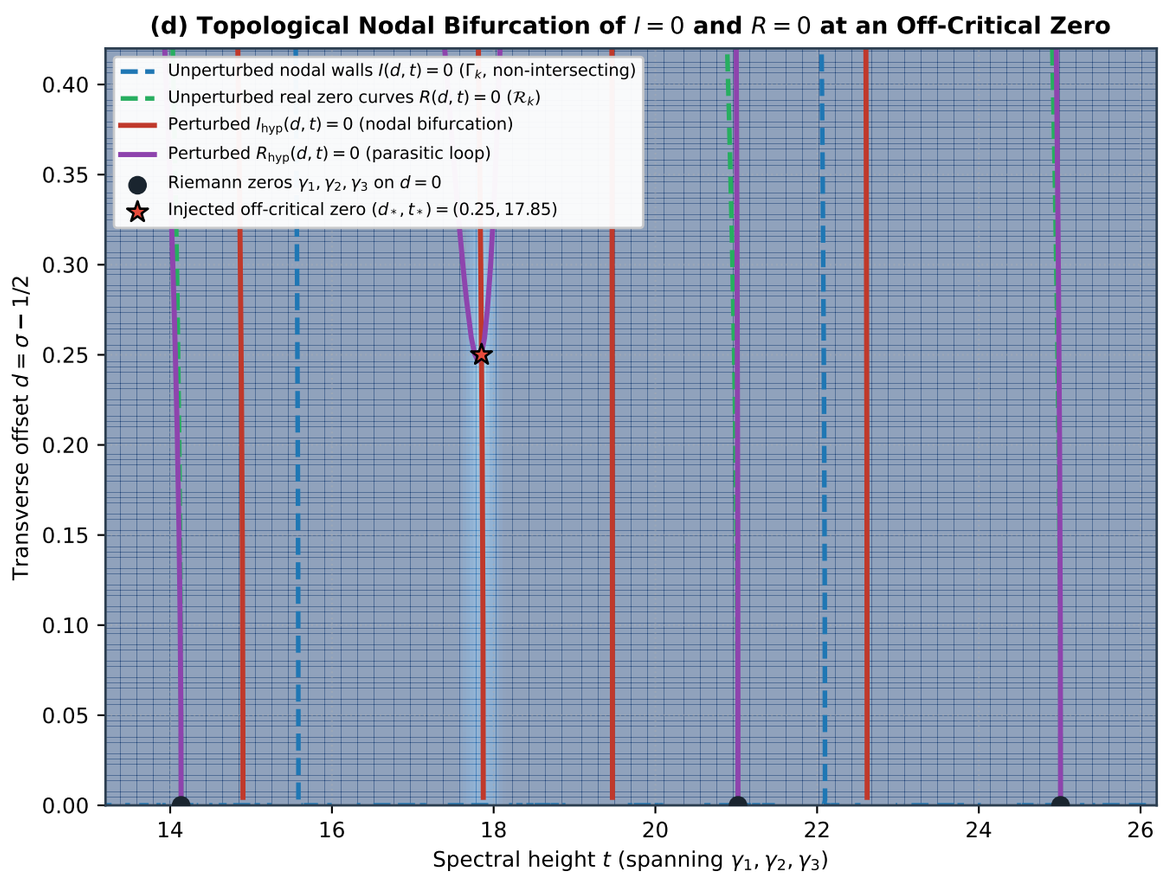

(a) Transverse Integral Law & Mandatory Rolle Turning Point $d_{\text{peak}} \in (0, d_*)$
True Riemann $|\widehat{\Xi}(t_* - \mathrm{i}d)|^2 = |\widehat{\Xi}(t_*)|^2 + 2\int_0^d u\widehat{W}\,\mathrm{d}u$
Hypothetical off-critical zero $3 \times |\widehat{\Xi}_{\text{hyp}}(t_* - \mathrm{i}d)|^2$ $(d_* = 0.25)$
Scaled derivative $0.025 \times \widehat{W}_{\text{hyp}}(d, t_*)$ (zero at d_{peak})
Positive build-up $\int_0^{d_{\text{peak}}} 2u\widehat{W}\,\mathrm{d}u > 0$ $(d_{\text{peak}} = 0.1342)$
Mandatory negative Rolle impulse $\int_{d_{\text{peak}}}^{d_*} 2u\widehat{W}\,\mathrm{d}u = -E(d_{\text{peak}}) < 0$
Normalized energy $|\widehat{\Xi}(t_* - \mathrm{i}d)|^2$
Transverse offset $d = \sigma - 1/2$


(b) Complementary Pure-Square Phase-Locking ($\widehat{\Xi}(t)^2$ at τ_k vs. $d^2(\widehat{\Xi}'(t))^2$ at γ_k)
Ground-state square $\widehat{\Xi}(t)^2$ (maximal at chamber bases τ_k)
Slope square $d^2(\widehat{\Xi}'(t))^2$ (maximal at Riemann zeros γ_k)
Complementary pure-square sum $\widehat{Q}_0(d, t) = \widehat{\Xi}(t)^2 + d^2(\widehat{\Xi}'(t))^2 > 0$
Exact integrated energy $|\widehat{\Xi}(t - \mathrm{i}d)|^2 = \widehat{\Xi}(t)^2 + 2\int_0^d u\widehat{W}\,\mathrm{d}u$ $(d = 0.35)$
4th-order dipole expansion $\widehat{\Xi}(t)^2 + d^2\hat{c}_1(t) + \frac{1}{2}d^4\hat{c}_3(t)$
Archimedean-normalized energy $(d = 0.35)$
Spectral height t (zeros $\gamma_1, \ldots, \gamma_5$ and extrema $\tau_1, \ldots, \tau_4$)


(c) Continuum Certification $t \in [14, 1000]$ & Low-Pass Filter $\mathcal{F}_{\text{int}}(2dy) = \frac{\tanh(dy)}{dy}$
Normalized integral energy $\widehat{\hat{\mathcal{E}}}(d, t) = |\widehat{\Xi}(t - \mathrm{i}d)|^2/d^2$ $(d = 0.35)$
Pointwise parity-reduced Wronskian $\widehat{\widehat{W}}(d, t)$ $(d = 0.35)$
Pure-square ratio $\mathcal{R}_{\text{sq}}(d, t) = |\Xi(t - \mathrm{i}d)|^2/\mathcal{Q}_0(d, t)$ $(\min = 0.6726 > 0)$
Integral energy floor $\min\widehat{\hat{\mathcal{E}}} = 0.6372 > 0$ (mean $2.11 \times \widehat{\widehat{W}}$)
Pointwise Wronskian floor $\min\widehat{\widehat{W}} = 0.6255 > 0$
Magnitude on logarithmic scale $(d = 0.35)$
Spectral height $t \in [14, 1000]$ (10,000 grid points)


(d) Topological Nodal Bifurcation of $I = 0$ and $R = 0$ at an Off-Critical Zero
Unperturbed nodal walls $I(d, t) = 0$ (Γ_k, non-intersecting)
Unperturbed real zero curves $R(d, t) = 0$ ($\mathcal{R}_k$)
Perturbed $I_{\text{hyp}}(d, t) = 0$ (nodal bifurcation)
Perturbed $R_{\text{hyp}}(d, t) = 0$ (parasitic loop)
Riemann zeros $\gamma_1, \gamma_2, \gamma_3$ on $d = 0$
Injected off-critical zero $(d_*, t_*) = (0.25, 17.85)$
Transverse offset $d = \sigma - 1/2$
Spectral height t (spanning $\gamma_1, \gamma_2, \gamma_3$)